10 Hybrid / Multiple Inheritance Questions with Explanation

Example :

```
class University:
    def __init__(self, University_Name, City, Number_Of_Colleges, **kwargs):
        super().__init__(**kwargs)
        self.University_Name = University_Name
        self.City = City
        self.Number_Of_Colleges = Number_Of_Colleges

class College(University):
    def __init__(self, College_Name, **kwargs):
        super().__init__(**kwargs)
        self.College_Name = College_Name

class School(University):
    def __init__(self, School_Name, **kwargs):
        super().__init__(**kwargs)
        self.School_Name = School_Nam

class Student(College, School):
    def __init__(
        self,
        University_Name,
        City,
        Number_Of_Colleges,
        College_Name,
        School_Name,
        Student_Name
    ):
```

```
        super().__init__(
            University_Name=University_Name,
            City=City,
            Number_Of_Colleges=Number_Of_Colleges,
            College_Name=College_Name,
            School_Name=School_Name
        )
        self.Student_Name = Student_Name

    def Display_Details(self):
        print("University Name :", self.University_Name)
        print("City :", self.City)
        print("Number Of Colleges :", self.Number_Of_Colleges)
        print("College Name :", self.College_Name)
        print("School Name :", self.School_Name)
        print("Student Name :", self.Student_Name)

# Create Object
Std1 = Student(
    "Mumbai University",
    "Mumbai",
    850,
    "KJ Somaiya College",
    "Somaiya School",
    "Sumit"
)

Std1.Display_Details()
```

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1. Company → HR + IT → Employee

Structure

```
    Company
    /    \
  HR      IT
    \    /
    Employee
```

Attributes

Company

- Company_Name
- Company_City

HR

- HR_Manager

IT

- Programming_Language

Employee

- Employee_Name

Explanation

The Employee class should inherit data from both HR and IT. Since HR and IT inherit from Company, the employee will have access to all parent attributes.

Concepts

Hybrid Inheritance
Multiple Inheritance
super()

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2. Hospital → Doctor + Nurse → Patient

Structure

```
     Hospital
     /      \
 Doctor    Nurse
     \      /
     Patient
```

Attributes

Hospital

- Hospital_Name
- Location

Doctor

- Doctor_Name

Nurse

- Nurse_Name

Patient

- Patient_Name

Explanation

The patient should have access to:

- Hospital information
- Doctor information
- Nurse information
- Patient information

Display everything together.

Concepts

Multiple Inheritance
Constructor Chaining

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3. University → College + School → Student

Structure

```
      University
      /      \
  College    School
      \      /
       Student
```

Attributes

University

- University_Name

- City

College

- College_Name

School

- School_Name

Student

- Student_Name

Explanation

The student belongs to both the college and school system and should inherit all information.

Concepts

Hybrid Inheritance
super()

---

4. Bank → Savings + Loan → Customer

Structure

```
      Bank
    /    \
 Savings   Loan
    \    /
   Customer
```

Attributes

Bank

- Bank_Name
- Branch

Savings

- Savings_Account_No

Loan

- Loan_Amount

Customer

- Customer_Name

Explanation

The customer has:

- Bank details
- Savings account details
- Loan details

Display complete customer information.

Concepts

Multiple Inheritance
Real-world Banking Example

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5. Vehicle → Car + Bike → Showroom

Structure

```
    Vehicle
    /   \
  Car    Bike
    \   /
   Showroom
```

Attributes

Vehicle

- Brand
- Model

Car

- Car_Type

Bike

- Bike_Type

Showroom

- Owner_Name

Explanation

The showroom stores information about both cars and bikes.

Concepts

Hybrid Inheritance
Parent Attribute Sharing

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6. School → Teacher + Library → Student

Structure

```
        School
       /      \
  Teacher    Library
       \      /
       Student
```

Attributes

School

- School_Name
- City

Teacher

- Subject

Library

- Book_Count

Student

- Student_Name

Explanation

A student belongs to a school, studies a subject, and uses the library.

Concepts

Multiple Inheritance
Constructor Chaining

---

7. ECommerce → Product + Seller → Customer

Structure

```
        ECommerce
        /      \
   Product    Seller
```

```
  \     /
  Customer
```

Attributes

ECommerce

- Website_Name
- Country

Product

- Product_Name

Seller

- Seller_Name

Customer

- Customer_Name

Explanation

The customer should see:

- Website details
- Product details
- Seller details

Concepts

Hybrid Inheritance
Real-world E-Commerce System

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8. CricketAcademy → Coach + Ground → Player

Structure

```
CricketAcademy
  /     \
Coach    Ground
  \     /
   Player
```

Attributes

CricketAcademy

- Academy_Name

- Location

Coach

- Coach_Name

Ground

- Ground_Name

Player

- Player_Name

Explanation

The player should inherit:

- Academy details
- Coach details
- Ground details

Concepts

Multiple Inheritance
Sports Management Example

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9. Abstract Bank → ATM + MobileBanking → User

Structure

```
     Bank (Abstract)
     /         \
   ATM     MobileBanking
     \         /
       User
```

Attributes

Bank

- Bank_Name
- Branch

Abstract Method

Show_Bank_Details()

Additional Attributes

ATM

- ATM_ID

MobileBanking

- Mobile_Number

User

- User_Name

Explanation

The User class must:

1. Implement the abstract method.
2. Access ATM details.
3. Access Mobile Banking details.

Concepts

Abstraction
Multiple Inheritance
Method Overriding

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10. Abstract University → College + Hostel → Student

Structure

```
     University (Abstract)
      /          \
  College       Hostel
      \          /
       Student
```

Attributes

University

- University_Name
- City

Abstract Method

Show_University()

Additional Attributes

College

- College_Name

Hostel

- Hostel_Name

Student

- Student_Name

Explanation

The student must:

- Implement the abstract method.
- Inherit college and hostel information.
- Display all details.

Concepts

Abstraction
Hybrid Inheritance
super()
Constructor Chaining

Interview Tip

When solving these questions:

Step 1

Draw the inheritance diagram.

Step 2

Identify which attributes belong to each class.

Step 3

Use:

super().__init__(...)

for constructor chaining.

Step 4

Create a display method in the final child class.

Step 5

Print all inherited attributes.

Easy Memory Trick

Single Inheritance
A → B

Multiple Inheritance
A + B → C

Multilevel Inheritance
A → B → C

Hybrid Inheritance
Combination of Multiple + Hierarchical Inheritance